



Improved prediction of biomass gasification models through machine learning

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ABSTRACT

Gasification of lignocellulosic biomass can be used to produce syngas used as a biorefinery feedstock. To facilitate the commercialisation of the gasification process, models are used to predict the outputs, simulate the impacts of irregular circumstances, and analyse process feasibility. This paper presents a hybrid model combining Aspen Plus and machine learning (ML) algorithms to enhance the prediction of gasification outputs. A base case gasification process flowsheet simulation was implemented in Aspen Plus based on assumed thermodynamic equilibrium conditions which can lead to inaccurate results. To address this, six ML algorithms were applied to collected experimental data and analysed for accuracy and efficiency. The feature importance, accuracy improvement, and the effect of implementing the ML predictions in the gasification block on the rest of the flowsheet were investigated. This paper emphasises the need of higher accuracy models and the great potential of ML approaches to offer high accurate predictions.

1. Introduction

Renewable and sustainable energy resources have gained prominence in recent research as a solution to address increasing greenhouse gas emissions that contribute to global warming. Using lignocellulosic biomass to produce biofuels offers a promising alternative to fossil fuels due to the sustainability and renewability of biofuels (Tursi, 2019). Thermochemical conversion technologies, such as gasification, are used to convert lignocellulosic biomass into biofuels and biochemicals due to their efficiency and high yield (Maity, 2015). Gasification is a partial oxidation process in which the lignocellulosic feedstocks break down into a gaseous, liquid, and solid products at high temperatures, ranging from 600 to 1000 °C, using different gasification agents such as air, steam, oxygen, or their mixtures. The gaseous products resulting from gasification are called synthesis gas (syngas), principally composed of CO and H₂. CO₂ and CH₄ in varied compositions and undesired gases such as H₂S are also present (Ibarra-Gonzalez and Rong, 2019). Syngas

can serve as a feedstock for producing liquid hydrocarbon fuels, methanol, and hydrogen or be used directly as fuel for internal combustion engines (Brown, 2015). An undesired liquid product, known as tar, consisting of high molecular weight compounds may also be produced (Serrano and Castelló, 2020). Additionally, a solid product, called char, is generated as a byproduct, mainly consisting of carbon and ash. Unlike tar, char is considered a valuable product, finding applications in soil amendment, wastewater treatment, carbon sequestration, and energy production (Khiari et al., 2019). The desired product ratio can be controlled through the choice of gasification agents. For instance, steam gasification produces a higher H₂/CO ratio than air gasification because the high partial pressure of H₂O offers higher reactivity of water-gas shift and methane reforming reactions, resulting in increased hydrogen production (Kumar et al., 2009).

This research aims to develop a commercial-scale model for the gasification of lignocellulosic feedstock. Thus, process flowsheet simulation has been selected as the main modeling method due to its ability

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to develop detailed plant flowsheets and enable plant-wide economic and sensitivity analysis. Several models simulating lignocellulosic gasification processes can be found in literature. Past research aimed to use models to conduct energy, technoeconomic and sensitivity analyses (Patel et al., 2016) and optimize and investigate integration opportunities with power production or other biomass valorization processes (Ibarra-Gonzalez and Rong, 2019). Lignocellulosic biomass structure is complex and this translates to the complexity of the gasification process. To simplify the models, different assumptions were made in previous research. A common assumption implemented in the literature is that gasification occurs in thermodynamic equilibrium, where yields are calculated by progressing reactions that minimize overall Gibbs Free Energy change (Safarian, Unnþórsson, et al., 2019). Other common assumptions include the omission of tar formation (Acar and Böke, 2019; Han et al., 2017; Hussain et al., 2018; Safarian, Richter, et al., 2019; Zaman and Ghosh, 2021) and the application of ideal gas properties for gas phase operations (Begum et al., 2014; Benalcázar et al., 2017; Dhanavath et al., 2018; Mehrpooya et al., 2018; Safarian, Richter, et al., 2019; Zaman and Ghosh, 2021). It is notable that in most cases, the predicted output is not well matched with experimental or pilot plant data (Dhanavath et al., 2018; Hussain et al., 2018; Pati et al., 2020; Yong and Abdul Rasid, 2021; Zhai et al., 2016). In cases that assume thermodynamic equilibrium, CH₄ is significantly underestimated. This can be attributed to the modelling of the consumption of CH₄ at equilibrium conditions in the steam reforming reaction (Acar and Böke, 2019; Han et al., 2017; Petersen et al., 2015; Safarian, Richter, et al., 2019; Tauqir et al., 2019; Zaman and Ghosh, 2021). Empirical data shows higher CH₄ yield values of syngas, which does not match the values predicted by current Aspen models. Tar is also considered non-equilibrium product and it is not accurately predicted in equilibrium models and the complexity of its structure makes it problematic to be modeled in Aspen Plus (Safarian, Unnþórsson, et al., 2019).

One approach to enhance the accuracy of gasification process flowsheet simulation models involves integrating non-equilibrium kinetic data. Previous research has shown that this approach improves prediction accuracy (Pauls et al., 2016; Puig-Gamero et al., 2021) and further extend this improvement to include tar and char in the analysis (Pauls et al., 2016). While kinetic models hold great potential to replace the thermodynamic equilibrium approach, they are not without limitations. Kinetic data may still be incomplete in some cases due to the complexity of the process. It is also unclear if kinetic models will be accurate under different operating conditions and with various biomass species. For example, multistep kinetic model is less accurate for operations conducted from 600 to 800 °C in pyrolysis process (Chen et al., 2020). Moreover, lumped type kinetic models are applicable to limited range of data, and it is less effective when determining tar and char yields. Distributed kinetic models are also restricted as they are specific for each feedstock (Vikram et al., 2021). Kinetic models can be also overly complex when they are integrated with mass transfer relations (Papari and Hawboldt, 2015).

Another method investigated in the literature involves employing machine learning (ML) algorithms to predict gasification output composition and properties. In many cases, ML effectively handles the underlying correlations between input and output data, even without prior knowledge of the physical and chemical description of gasification reactions. The ML based outcomes have been shown to be at par with analytical and mechanistic approaches (Schweidtmann et al., 2021; Vikram et al., 2021). Developing a reliable machine learning model requires significant amounts of data for training and testing (Baruah and Baruah, 2014). Various machine learning techniques have been applied to predict gasification output variables, such as artificial neural network (ANN) (Baruah et al., 2017; Kargbo et al., 2021; Li, et al., 2018a, 2018b; Serrano and Castelló, 2020), and tree-based models (Elmaz et al., 2020; Mutlu and Yucel, 2018). ANN is a ML model inspired by the structure and functioning of the human brain, while tree-based models make predictions by partitioning the input data into tree-like structures. The

coefficient of determination (R^2) of machine learning approaches often exceeds 0.90, showing great potential to overcome poor accuracy. ML is not limited to predicting output composition, it can also be applied for optimization purposes, such as maximizing gasification rate and conversion (Kargbo et al., 2021; Nassef et al., 2020). However, due to the vast diversity of biomass species and operating conditions, there are still knowledge gaps to address in exploring machine learning's potential to predict gasification output.

The application of ML algorithms to improve existing process flowsheet simulation models is an emerging research interest. While minimisation of Gibbs Free Energy change in the process has been widely implemented in gasification literature, we propose a ML model to overcome inaccuracies identified in mechanistic models. Hybrid ML-Aspen models have been previously reported, but these models were utilising Aspen Plus to only generate data for ML model training and development.

Therefore, this work represents the first hybrid Aspen Plus-ML gasification prediction model developed for biomass gasification application, to the best of our knowledge. It fuses the traditional first principles and data-driven models to utilise the advantages of both methods namely, modelling versatility and detailed simulation offered by Aspen Plus and prediction accuracy of ML algorithms. The air-steam gasification system was modelled as the base case in Aspen Plus. ML methods were developed in Python and integrated with Aspen Plus to provide predicted data for the entire process flowsheet. Data used to develop the ML models was collected and processed using quality criteria. A total of 798 gas composition, tar and gas yield datapoints were collected to develop the ML models. Developing accurate predictions of the gasification process can simulate the biofuel production process more accurately. The results of the process flowsheet simulation, specifically, gaseous, liquid, and solid yields along with heat duties and CO₂ emissions results using the ML model were compared to the base case process flowsheet simulation in Aspen to demonstrate the improvement in prediction.

2. Methods

2.1. Simulation model process description

Aspen Plus V12.0 was used to develop the process flowsheet simulation of lignocellulosic biomass gasification. The flowsheet consists of four main sections, namely drying, gasification, gas cleaning, and Fischer-Tropsch (FT) synthesis. The "UNIQUAC" activity coefficient model was used to handle thermodynamic calculations of most of the flowsheet except gas cleaning in which the electrolyte non-random two liquids (eNRTL) activity model was applied as suggested in the literature (Abdul M Petersen, 2014). UNIQUAC is the most efficient property method to estimate thermodynamic, binary interaction, and transport properties in biomass processing models (Onarheim et al., 2014). Biomass was depicted in Aspen Plus as non-conventional solid (MIXNC stream class) and its enthalpy and density were estimated using HCOALGEN and DCOALIGT predefined models. The detailed composition of biomass was defined by its proximate and ultimate analysis. To evaluate the prediction accuracy of the model, the process flowsheet simulation was developed based on the operating conditions and biomass characteristics of the steam gasification of the work of (Loha et al., 2011) as listed in Table 1.

The lignocellulosic gasification process flow diagram developed in Aspen Plus is shown in the Aspen Plus section of Fig. 1. The modelled process is described in the following paragraphs. The first section is the drying section where the biomass is dried by hot air in the "Drying" block to reduce its moisture content to the desired level depending on the case simulated. Then, the dried biomass enters the gasification section which starts with an "RYIELD" reactor. The aim for this step was to convert biomass from a nonconventional component to its elemental composition as conventional components based on proximate and

Table 1

Biomass characteristics and gasification conditions used in the base case process flowsheet simulation adapted from a random datapoint from (Loha et al., 2011).

Biomass characteristics (sugarcane bagasse)	
Proximate analysis (wt% dry basis)	
Volatile matter	81.7
Fixed carbon	14.45
Ash	3.85
Moisture content	13.05
Ultimate analysis (wt% dry basis)	
C	46.1
H	5.26
O	44.25
N	0.19
S	4.2
Operating conditions and gasification agent quantity	
Gasification temperature (°C)	750
Steam/biomass (weight basis)	1.25

ultimate analysis so the Aspen Plus can carry out the phase and chemical equilibrium calculations. A calculator module was included to automate the “RYIELD” calculations for any biomass feedstock type and composition introduced instead of requiring manual adjustment for each case. After that, the biomass is fed to the gasification reactor “RGIBBS” which predicts the gaseous, liquid, and solid output based on Gibbs Free Energy minimisation method assuming thermodynamic equilibrium conditions and infinity reaction time (Anon., Aspen Technology, 2000). The gasification agents such as steam, air, and mixture of both are also fed to the “RGIBBS” reactor. Steam and air flow rates were determined based on steam/biomass ratio and equivalence ratio data, respectively. The thermodynamic equilibrium approach is implemented as a base case as it is the conventional method used to carry out the gasification reactions process in process flowsheet simulations. The products of the gasification are fed into a solid separator to eliminate any residual solids. Then, the gaseous and tar products are fed into a tar reformer to convert CH₄ and tar components into CO₂ and H₂.

After that, most of the H₂S content of the syngas must be removed below 4 ppm to meet downstream processing requirements. The technology used in the present work is amine absorption by mono-diethanolamine (MDEA). Syngas is contacted with MDEA in the absorber to absorb the H₂S. The amine will then be regenerated using a stripper where H₂S is removed and the MDEA is recycled back to the absorber. After conditioning the syngas, it is sent to the FT synthesis unit to produce liquid hydrocarbons. Finally, the hydrocarbons are distilled and fractionated into their pure components to be sold separately. All

operating conditions and design variables used in this flowsheet were adapted from a US National Renewable Energy Laboratory (NREL) report (Swanson et al., 2010) and any missing variables were obtained from the literature. These variables are detailed in Table S3 in the supplementary data.

2.2. Machine learning process

To improve the prediction of gasification product yields and gas composition, ML algorithms were used to replace the “RGIBBS” reactor, which was typically used to simulate gasification reactions. The performance of six different ML algorithms namely, linear regression, ridge regression, ANN, regression trees, random forest, and Gradient Boosting ensemble tree model (XGBoost) was investigated through training and testing using collected empirical data to develop a machine learning model. Each ML method has its unique characteristics and parameters that were investigated to obtain the best performance. The prediction accuracy was evaluated, and the most accurate method was integrated with the process flowsheet simulation. The Scikit-learn (sklearn) package in Python 3.9.12 was used for data splitting and ML model development. The detailed methodology of dataset gathering and preprocessing followed by model evaluation measures is described in the next subsections. Then, the development of linear ML models (linear and ridge regression), ANN and tree-based models (regression trees, random forest, and XGBoost) is discussed.

2.2.1. Dataset gathering

In this work, empirical data (including laboratory and pilot plant experimental data) reported in the literature was identified to be reliable due to their accessibility and availability. Platforms such as Google Scholar and databases including Scopus and Web of Science have been utilised to gather the required data from a total of 51 various research articles to develop the ML models. Keywords including lignocellulosic biomass gasification, steam gasification, and air gasification of biomass were used in the data search. There was no minimum year of publication selected for this study to maximise the number of collected data points. More details about the data used to develop the ML models are found in the Supplementary Material. After an extensive review of the literature, it was determined that the variables that were typically manipulated in the empirical data were biomass characteristics, gasification temperature and gasification agents’ quantity, which was specified as either equivalence ratio (ER), steam to biomass ratio (S/B) or both, depending on the system. The corresponding dependent variables collected included syngas composition (CO, CO₂, H₂, CH₄), tar, and gas yield. To

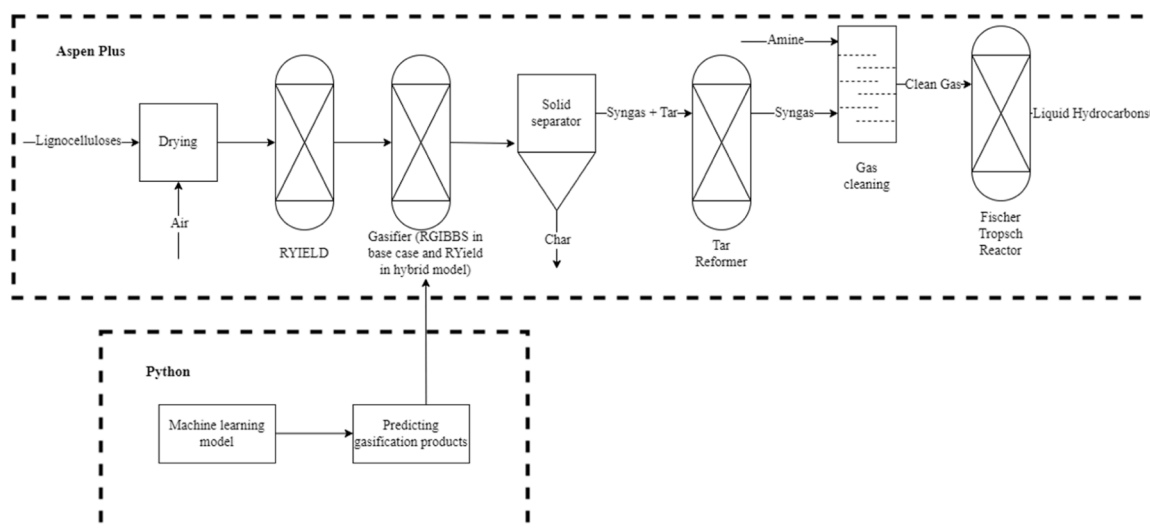


Fig. 1. The hybrid model structure.

ensure the quality of the data, three main criteria were considered during collection, as follows:

- The quality of the publication journal. This was evaluated, considering the journal rank and its quartile placement based on SCImago Journal Rank (SCImago, 2024). Most of the data gathered are adapted from Q1 journals.
- Each datapoint must include all the features (input variables) required to construct the model.
- The data is obtained by more than one experimental trial which demonstrates its reproducibility and validity.

Table 2 presents the dataset size, detailed features and target variables gathered for the present study. These datapoints originate from studies involving diverse gasification configurations, including various reactor types and hydrodynamics, catalyst and bed materials, although these variables were not considered for the present analysis since these data were not consistently reported in the scanned literature.

2.2.2. Data preprocessing and model evaluation

The data gathered was processed to uncover any inconsistencies and prevent errors in prediction. To do this, the data distribution frequency was plotted to detect and eliminate outliers. Determining the most efficient representation of the data for modelling was also an essential step in preprocessing. For example, the data around the gasification agent can be expressed either as categorical data, labelled as “steam”, “air”, or “mixture” with the quantity of the agents as additional features, or as continuous data represented by their quantities only. When handling continuous data, neither one-hot encoding nor scaling techniques were applied to it. The variable types used in the modeling process were chosen to accurately represent the information associated with the data and to reduce complexity as much as practicable.

The data was then divided into training and testing sets, with an 85 % allocation for training and 15 % for testing. No separate data was reserved for validation because the 5-fold cross validation method was used during training and testing of each algorithm. The average root mean square error (RMSE) and coefficient of determination (R^2) were used as the error indicators in each model run and were calculated as follows (Simon Ascher et al., 2022):

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N}} \quad (1)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \bar{y})^2}{\sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (2)$$

Here, N is the number of datapoints, y_i is the actual value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the mean of the actual values.

2.2.3. Linear and ridge regression

The linear and ridge regression were the simplest techniques employed in this work. Both were used to establish linear correlation among the investigated input variables and the target variables. For implementation of these techniques to our data, their libraries were imported from the sklearn package. Ridge regression introduces a regularisation hyperparameter to adjust the model, for improving its accuracy and reducing overfitting compared to linear regression. In each run of ridge regression, the optimum alpha hyperparameter was determined by cross-validation.

2.2.4. Artificial neural network

The MLPRegressor (Multilayer Perceptron) function, imported from sklearn package and implemented in Python, was used to train and test the ANN model by feedforward neural network approach. The input, hidden, and output layers are connected by a series of weights and they are updated in each iteration by the backpropagation method. The training process stops when the error between the predicted and actual output value is minimised to a certain level (Sun et al., 2023). Various hyperparameters were optimised, including the number of hidden layers, number of neurons in the hidden layers, the type of activation function, and the maximum number of iterations to produce best possible accuracy with no overfitting. These hyperparameters were optimised using Grid Search method (i.e. creating a grid of possible values that tests different combinations of hyperparameters which returns the best possible fit). Different output layer configurations can be used in the ANN model. Therefore, the predictability of five different models was tested to predict gasification products. One multi output model that considered all gas constituents simultaneously, while four models predicted each gas constituent composition separately. The selection between multi and single output structure for each gas composition depends on R^2 value of the prediction.

2.2.5. Regression trees, random forest, and XGBoost

Sophisticated techniques from tree-based model family namely, regression trees, random forest, and XGBoost were employed in this work. Tree-based models are widely accepted supervised algorithms capable of handling non-linearity or collinearity in complex datasets. DecisionTreeRegressor, and RandomForestRegressor libraries were used in this study for regression trees and random forest models, respectively, and both were imported from the sklearn libraries. For the XGBoost regression technique, the XGBoost package was utilised for implementation in Python. The Random Forest model consists of individual regression trees operating independently, employing the bagging ensemble technique to yield the final output determined by the majority decision of the trees. This process improves the prediction performance by controlling overfitting on training data. The XGBoost method constructs a sequential ensemble of tree models, each collaborating to enhance one another and collectively determine the ultimate output. It utilises second-order Taylor series expansion of the loss function and includes a regularisation term to obtain an optimum solution (Sun et al., 2023). Hyperparameters were optimised using Grid Search method before generating the final output and reduce the overfitting. The aim was to generate training and testing R^2 results that are close to each other to ensure that the high accuracy is not resulted from overfitting the

Table 2
Variables, data range, and data size used to develop ML models.

Features		
Variable	Unit	Experimental data range
Volatile matter (VM)	wt%	52.56 – 83.70
Fixed carbon (FC)	wt%	6.86 – 26.45
Moisture Content (MC)	wt%	4.56 – 27.00
Ash	wt%	0.01 – 19.52
Carbon (C)	wt%	43.30 – 58.34
Hydrogen (H)	wt%	3.57 – 8.66
Oxygen (O)	wt%	31.01 – 51.83
Nitrogen (N)	wt%	0.01 – 6.55
Sulfur (S)	wt%	0 – 4.20
T emperature	°C	599.00 – 1108.00
Steam to biomass ratio	Weight basis	0 – 4.70
Equivalence ratio	Weight basis	0 – 0.50
Targets		
CO composition	Vol%	7.37 – 50.56
CO ₂ composition	Vol%	5.00 – 59.03
H ₂ composition	Vol%	6.36 – 65.66
CH ₄ composition	Vol%	0.43 – 22.00
Gas yield	m ³ /kg biomass dry basis	0.26 – 3.81
Tar	g/m ³ gas	0.46 – 20.75
Number of target variables datapoints		
Gas composition data		417
Gas yield		269
Tar data		112

training data. For regression trees and random forest, the common hyperparameters adjusted were minimum number of samples per leaf, maximum depth of a tree, and maximum number of leaf nodes, while the number of estimators hyperparameter was unique for random forest. The hyperparameters tuned for XGBoost were maximum depth of a tree, ratio of features used, ratio of the training instances used, minimum sum of instance weight needed in a leaf, learning rate, and minimum loss reduction. More details about the hyperparameters value ranges tested in this work are available in Table 3.

2.3. Integrating machine learning with process flowsheet simulation models

Machine learning predictions were integrated with the developed Aspen Plus flowsheet. The RGIBBS block (gasifier), shown in Fig. 1, where the gasification reactions take place, had been replaced with RYIELD reactor type block to incorporate the output generated from ML predictions. Some output components, such as water and char, were not consistently reported in the literature, so these were calculated using mass balance correlations to ensure convergence in the process flowsheet simulation. Table 4 breaks down the outputs considered in this study and the method used to deliver the results. The model can be run via Python or Aspen flowsheet to generate comprehensive mass and energy balance results for the entire flowsheet.

Testing data (63 data points) were used to conduct the comparison between conventional and ML models. For this analysis only, the order of use of testing data was fixed in ML models and did not follow random distribution in each iteration, unlike other analyses conducted in this research.

3. Results and discussions

3.1. Simulation model

Table 5 compares the calculated values of the process flowsheet simulation with the actual values reported by (Loha et al., 2011). Results in Table 5 indicate deviations between these values except for CO₂. CO estimations are slightly underestimated in this work while H₂ compositions is slightly overestimated. It is important to note that there is almost no estimation of CH₄ using the Aspen model, and this behaviour is consistent with the literature (Acar and Böke, 2019; Han et al., 2017; Petersen et al., 2015; Safarian, Richter, et al., 2019; Tauqir et al., 2019;

Table 3

The range of values of the hyperparameters tuned for ML approach.

ML approach	Hyperparameter	Range of manipulation	Step size
Regression trees	Minimum number of samples per leaf	4–8	1
	Maximum depth of a tree	2–8	1
	Maximum number of leaf nodes	5–20	5
	Minimum number of samples per leaf	2–4	1
Random forest	Maximum depth of a tree	2–5	1
	Maximum number of leaf nodes	15–20	1
	Number of estimators	20–90	10
	Maximum depth of a tree	3–6	1
	Ratio of features used	0.4–0.7	0.1
	Ratio of the training instances used	0.4–0.7	0.1
	Minimum sum of instance weight needed in a leaf	2–4	1
XGBoost	Learning rate	0.05–0.10	0.01
	Minimum loss reduction	0.5–5.0	0.5
	Maximum number of iterations	500–2000	500
	Activation function	Tanh, ReLU	–
	Number of hidden layers	1–2	1
ANN	Number of nodes in the hidden layer	5–100	5

Table 4

Outputs prediction and calculation methods for the gasifier, presented as yield reactor within the integrated flowsheet, along with their representations.

Component	Method	Note*
CO	Machine learning	Total gas yield is also predicted using machine learning approaches
CO ₂	Machine learning	
H ₂	Machine learning	
CH ₄	Machine learning	
Tar	Machine learning	Represented as benzene
Char	Mass balance	Calculated using carbon conversion efficiency (CCE) equation: $CCE = \frac{12\phi(CO + CO_2 + CH_4)}{22.4X_c} \cdot 100 \quad (3)$
Water	Mass balance	Represented as solid carbon Calculated using steam decomposition (SD) equation: $SD = \frac{V_{gs}(H_2 + 2 \cdot CH_4) \cdot \frac{18}{22.4}}{W_1 + W_2} \cdot 100 \quad (4)$
H ₂ S	Mass and mole balance	Assuming 100 % conversion of sulphur content of biomass by the following reaction: $H_2 + S \leftrightarrow H_2S \quad (5)$
N ₂	Mass balance	Inert
Ash	Mass balance	Inert and represented as Si

* ϕ is the volume of the produced gas per kg of biomass (gas yield), CO, CO₂, H₂ and CH₄ are the volume fractions of components of the product gas, X_c is the mass fraction of carbon in ultimate analysis, V_{gs} is dry product gas flow rate (Nm³/h), W_1 is steam flow rate, and W_2 is total moisture content of the feed.

Table 5

Comparison of the predictions of gaseous products by Aspen Plus flowsheet of this study and the work of (Loha et al., 2011) using temperature = 750 °C and S/B = 1.25.

Gas product compositions (vol%)*		
Gas component	RGIBBS in this work	(Loha et al., 2011)
CO	15.66	19.74
CO ₂	24.35	21.52
H ₂	59.14	54.17
CH ₄	0.84	4.57

* The computed gas compositions in this table are out of total flow rate of the gaseous products.

Zaman and Ghosh, 2021). The main reason for this result is the assumption of thermodynamic equilibrium, which is an ideal condition for the steam reforming reaction where methane is consumed. Table 6 shows that in the gasification cases that operate temperatures higher than 750 °C, Aspen Plus significantly underestimates CH₄ and in most cases the CH₄ composition values are <0.1 %. The assumption of thermodynamic equilibrium in the Aspen Plus RGibbs reactor has a significant impact on the estimation of the gas composition. Under this condition, the steam reforming reaction is dominant. The reaction can be expressed as the following:

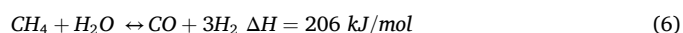


Table 6

Estimated CH₄ composition in Aspen Plus at various temperature ranges.

Temperature range	Percentage of CH ₄ predicted composition < 0.1	Percentage of CH ₄ predicted composition > 0.1
650 ≤ Temperature < 750	13.33 %	86.67 %
750 ≤ Temperature < 850	90.91 %	9.09 %
850 ≤ Temperature	100.00 %	0 %

The endothermic behaviour of this reaction indicates that increasing the temperature will lead to higher methane consumption and hydrogen production. This effect was also observed in the overestimation of H_2 yield in 61 out of 63 cases, as shown in Fig. 2(c).

This assumption can impact the rates of the reactions and force some reactions that favour equilibrium conditions to take place, such as steam reforming (Safarian, Unnþórsson, et al., 2019)

3.2. Machine learning approaches

3.2.1. Pre-processing results

Results reported in Table 7 show that the type of data can influence the machine learning model performance. Continuous data offers higher R^2 than categorical data. Continuous data was found better than categorical data for training and testing CO_2 yield data, followed by the fit of testing data of CH_4 , while the other R^2 values have slightly better or similar performance with categorical data. Therefore, continuous data has been selected to be used for the rest of the work.

3.2.2. Linear and ridge regression

The machine learning phase of this study was commenced by developing linear and ridge regression models, which are the simplest techniques among the selected algorithms. To maximise the R^2 value, the ridge regression model possible includes the consideration of an optimal alpha (α). Fig. 3 shows the R^2 values of the models for predicting the target variables. In both linear and ridge regression, the R^2 values fall below 0.60, indicating that these models failed to effectively fit the dataset and accurately predict the output, with the exception of H_2 . This observation implies that the correlation between input and output is too intricate for linear models to handle. Therefore, more advanced models such as regression trees, random forest, XGBoost, and ANN were employed for this study.

3.2.3. Artificial neural network

To achieve the highest accuracy performance from an ANN model, its

topological structure must be determined by computing RMSE for different network configurations. The network configuration that yields the lowest average RMSE is selected as the neural network to be used for further purposes. The parameters subject to tuning include the number of hidden layers, number of neurons in the hidden layers, and the transfer function. The chosen network configuration and their corresponding results are listed in Table 8. These results are based on random data distributions in each iteration and have been obtained using 5-fold cross validation. As shown in Table 8, the estimation of CO , CO_2 , and CH_4 models were more accurate using single output models while H_2 is better predicted using multioutput model. Therefore, the method that offers the highest R^2 when predict each component composition have been selected for comparison with other ML approaches, as illustrated in Fig. 3.

The model was capable of fitting the learning data for all gas compositions while testing results show promising performance and the lowest R^2 is 0.77 computed for CH_4 prediction while the peak R^2 found is 0.87 for both H_2 and CH_4 . In general, the results show promising efficiency of ANN to predict gas compositions using the optimized structure.

3.2.4. Tree-based models

Results of predicting CO , CO_2 , H_2 , and CH_4 using tree-based models (i.e., regression trees, random forest and XGBoost) are displayed in Fig. 3. It is evident that tree-based models offer superior prediction accuracy for all gas components than linear models. To maximise their accuracy (i.e., minimised RMSE) and minimise overfitting, various parameters such as maximum depth of a tree, maximum number of leaf nodes, and learning rate were adjusted. The hyperparameters ranges and step sizes are detailed in Table 3. The results in Fig. 3 demonstrated very close R^2 values for learning and testing data for each method, which emphasise that the high performance of ML approaches is not generated from overfitting. XGBoost stands out as the most capable among the other tree-based models for predicting the syngas compositions. Hence, the XGBoost model was chosen based on the best performance on optimised hyperparameters values. The results including error

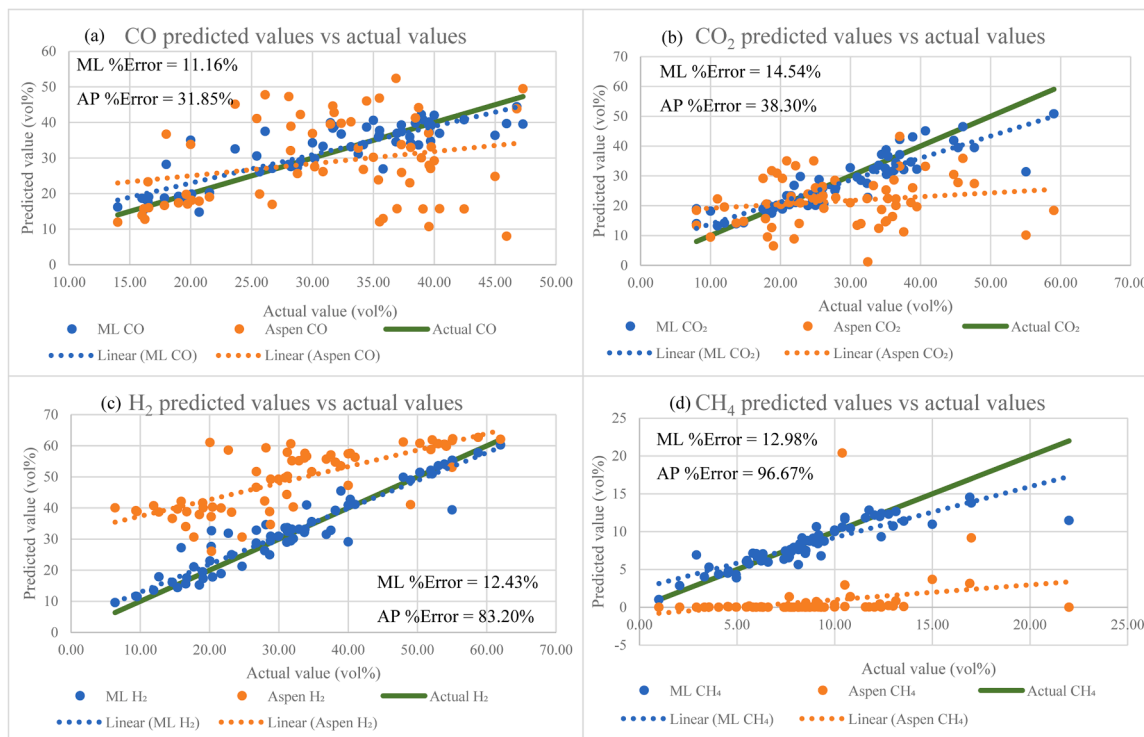


Fig. 2. Comparison between testing ML (XGBoost) and Aspen Plus (RGibbs) results for 63 different cases for predicting (a) CO, (b) CO₂, (c) H₂, and (d) CH₄ and comparison of linear fit between RGibbs (orange), and ML (blue) predictions with actual values (green).

Table 7

Performance of ML models using either categorical data and continuous data (C: Categorical data, N: Continuous data). Letter indicates the data type that resulted in the higher R^2 value.

Method	CO		CO ₂		H ₂		CH ₄	
	Learning	Testing	Learning	Testing	Learning	Testing	Learning	Testing
Linear Regression	C	N	N	N	N	N	C	C
Ridge Regression	C	N	N	N	N	N	C	C
Regression Trees	N	N	N	N	N	N	N	N
Random Forest	N	N	N	N	N	N	N	N
XGBoost	C	N	C	N	N	N	C	N
ANN	N	N	N	N	N	N	N	N

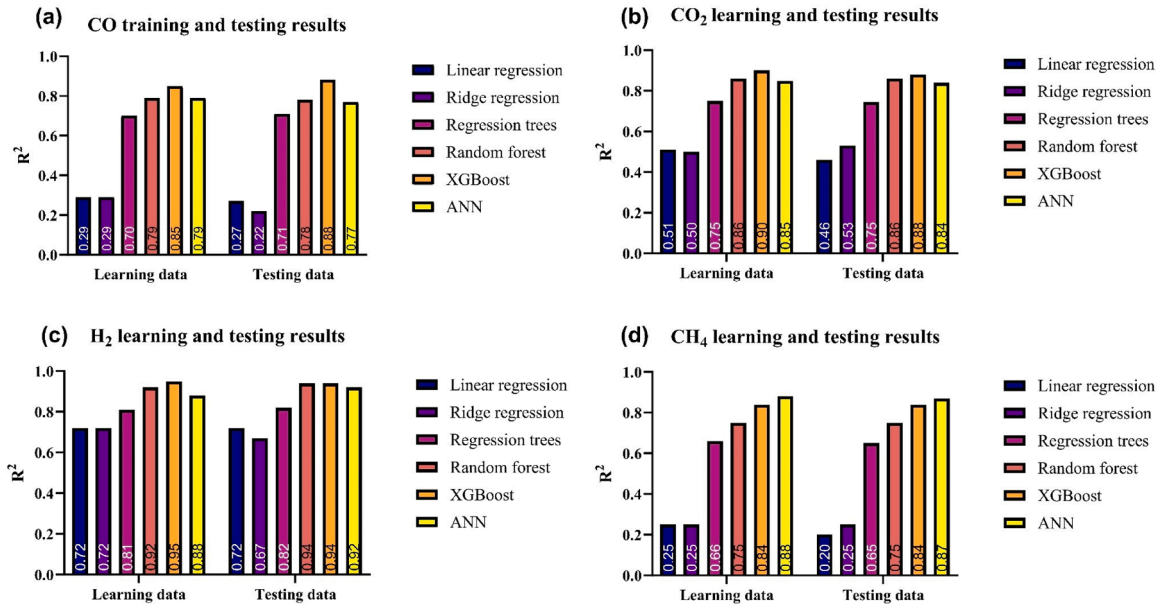


Fig. 3. Average R^2 values for six machine learning algorithms for predicting (a) CO, (b) CO₂, (c) H₂, and (d) CH₄.

Table 8

The optimised structure and results of single output models and all output model using ANN.

Characteristic	Specification
Number of input layer neurons	12
Number of hidden layer (s)	1
Number of hidden layer neurons	CO Model: 95, CO ₂ Model: 100, H ₂ Model: 100, CH ₄ Model: 65, All outputs model: 90
Number of output layer neurons	1 for single output models and 4 for multiple output model
Network type	Multi-layer perceptron
Transfer function	Tanh (for all cases)
Model solving method	Adam (stochastic gradient-based optimizer)
R^2 results based on 5-fold cross validation using the fine-tuned network configuration	
CO Single Output Model	0.77
CO ₂ Single Output Model	0.84
H ₂ Single Output Model	0.87
CH ₄ Single Output Model	0.87
CO Multioutput model	0.67
CO ₂ Multioutput model	0.78
H ₂ Multioutput model	0.92
CH ₄ Multioutput model	0.60

indicators are depicted in Fig. 4.

Additionally, a feature importance analysis was conducted in the XGBoost model, which measures the significance of individual features in predicting the target variable. The analysis calculates F score using

the “weight” metric, representing the number of times a feature is used to split the data across all trees in the model. Results in Fig. 5 show that temperature is the most important variable in the prediction of all gas compositions. This can be attributed to the known effect of temperature on altering H₂ and CO selectivity, as an increase in gasification temperature leads to an increase in syngas yield, in general, and H₂, in particular as a final product. This aligns well with some of the feature importance analyses conducted in the literature (S. Ascher et al., 2022; Safarian et al., 2021). Another observation from Fig. 5 was that the second most important variables are S/B and ER, which are quantities of gasification agents. It was proven in the literature that high S/B and ER affect the products and leads to an increase in H₂/CO ratio (Acar and Böke, 2019; Mansur et al., 2020) which explains the importance of these variables. This observation was identical to all gas components. Biomass characteristics were found to be less important than gasification conditions when predicting gas composition. For all output predictions, biomass characteristics have <50 % importance compared to gasification conditions, although the number of variables of biomass characteristics was more than that of gasification conditions. This result diverges from some ML studies that indicated that biomass characteristics especially nitrogen and ash content can heavily influence the product yields (J. Li et al., 2023; Yang et al., 2023). The variations between this work and these results are reasonable due to the differences in datasets used, ML methods used among different studies (H. Li et al., 2023), and feature importance measures employed (S. Ascher et al., 2022). Generally, the results of feature importance analysis cannot be used to predict outputs and applied to industrial process model. It only functions as an identifier of the features of the ML model used to

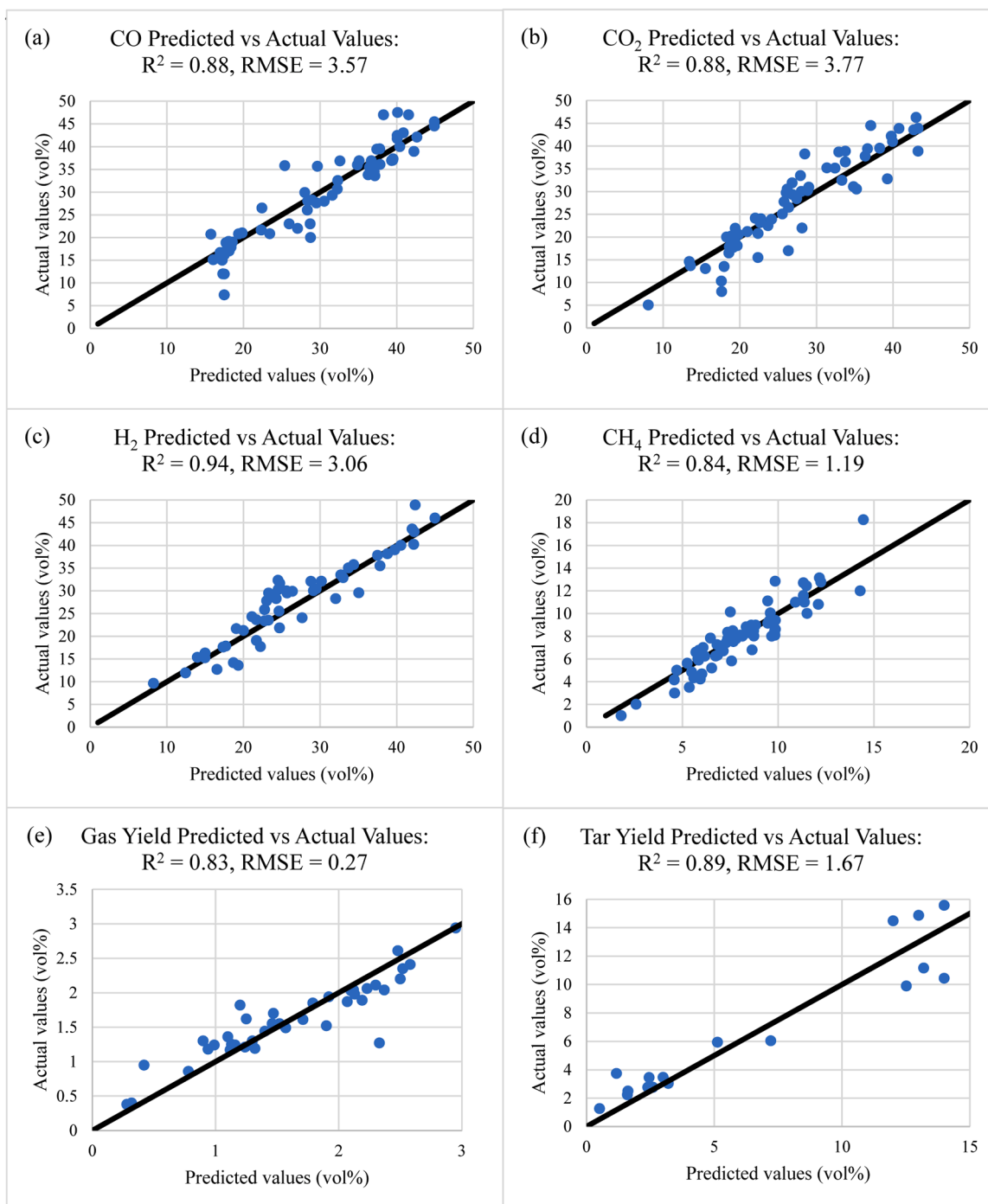


Fig. 4. Comparison between actual and predicted values of (a) CO, (b) CO₂, (c) H₂, and (d) CH₄ compositions, (e) gas yield, and (f) tar yield using XGBoost.

estimate the output products and is merely descriptive of the relationships between features and targets providing a starting point for analysing the model. Incorporating more datasets, adding more features, and performing further in-depth analysis can achieve a more comprehensive understanding of the correlations between input and output variables, but is limited by resources and computational power, so a balance between benefits and trade-offs should be considered.

Overall, as shown in Fig. 3, the results obtained from the ANN algorithm are competitive with XGBoost. The XGBoost results show superior performance when predicting CO and CH₄ composition, while the ANN model shows superior capability in predicting CO₂ yield. In the case of H₂ yield prediction, both models show similar behavior, especially for the values of R^2 for testing data. Therefore, due to the higher

efficiency and improved result comprehensibility achieved with XGBoost, it was selected as the algorithm for predicting the gas yields in the Aspen Plus flowsheet. This conclusion agrees well with the findings of Ayub et al. (2023), who conducted a similar investigation in a hydrothermal gasification system.

XGBoost was used to estimate the gas and tar yields employing 269 and 112 datapoints, respectively. Following a procedure similar to developing the gas composition models, the optimum values of hyper-parameters were determined through 5-fold cross validation for both runs to enhance prediction accuracy and reduce overfitting ($R^2 \sim 1$). Reasonable R^2 values (>0.80) have been achieved as shown in Fig. 4(e) and (f). The feature importance results for gas yield and tar exhibited a similar trend to that of gas compositions, particularly for the three most

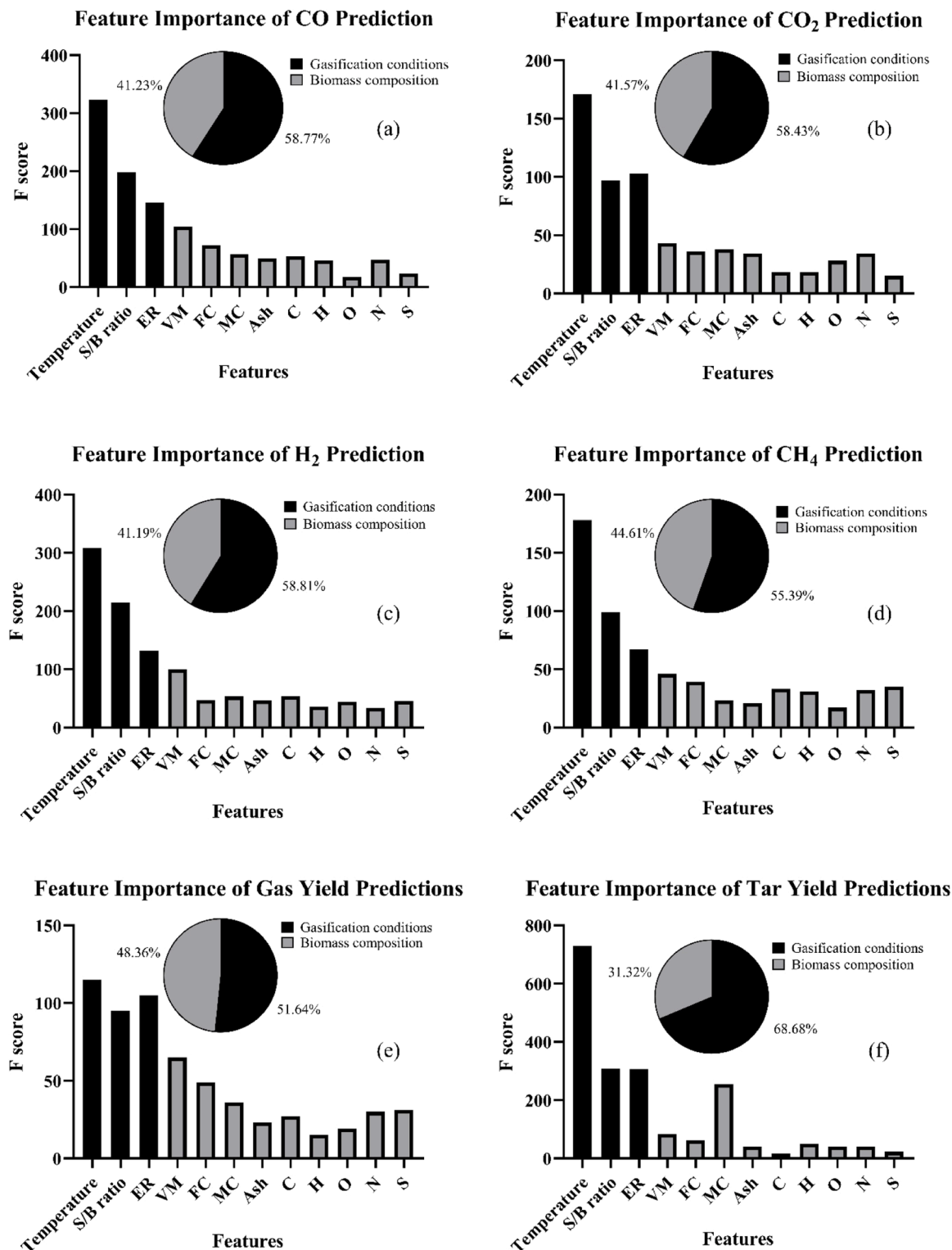


Fig. 5. Feature importance analysis (bar graph) and percentage of F score of gasification conditions and biomass composition (pie graph) of (a) CO, (b) CO₂, (c) H₂, (d) CH₄, (e) total gas yield, and (f) tar yield XGBoost predictive models.

significant features, as illustrated in Fig. 5(e) and (f).

3.3. Comparison between machine learning and Aspen Plus performance

To validate that ML algorithms provide an effective alternative to the minimisation of Gibbs Free Energy approach used in Aspen Plus when modeling the gasification conversion step, the performance of the

methods has been compared by assessing accuracy, particularly, error percentage calculated using the following equation:

$$\%Error = \frac{|\text{Predicted value} - \text{actual value}|}{\text{Actual value}} \times 100 \quad (7)$$

Here, the predicted value refers to the gas composition calculated by either Aspen Plus or XGBoost. The average error for both models for

each gas composition and the prediction distribution are presented in Fig. 2. The results indicate that XGBoost outperforms Aspen Plus projections, especially in the case of H₂ and CH₄. While Aspen Plus predictions for CO and CO₂ exhibit lower errors than H₂ and CH₄, they were more accurately estimated using XGBoost (see Fig. 2(a) and (b)) with an average error < 11 %. These results emphasise the capability of XGBoost to improve the prediction accuracy of gasification products. The few inaccurate estimates by the ML model can be attributed to the lack of other input data, such as the catalyst type and residence time, as some of the derived data are from studies that investigate the influence of changing catalyst types on gasification product compositions. The results generated by the XGBoost model, along with the required complementary mass balance, have been integrated into the Aspen Plus flowsheet.

Using different predictive approaches can have an impact on the model's results, thus the analysis between approaches has focused on the accuracy of consequential results. Table 9 presents the results of using either XGBoost or conventional Gibbs Free Energy minimisation calculation for gasification in the flowsheet, using the design basis of (Loha et al., 2011). Similar design variables and operating conditions were maintained for both cases during this comparison. Under thermodynamic equilibrium conditions and indefinite residence time, more gaseous products are generated compared to the XGBoost output, which is based on experimental data with limited reaction time. There are also disparities in the amount of char and tar produced by both approaches. In the case of ML method, the estimates of tar and char are higher than those from the conventional approach. These results directly impact the heat duty of the gasifiers, tar cracking reactor and the reboiler duty of amine regenerator column located in the gas cleaning section (see Table 9). A basic environmental assessment factor, namely CO₂ emissions, was calculated to assess any impacts of the results on the environmental analysis. The outcome shows that the conventional approach led to lower estimated CO₂ emissions compared to the ML approach. The effect of using different predictive approaches on the economic analysis results has not been investigated in this study, but it is expected to yield different results for each method since it is based on the mass and energy balance of the flowsheet (Farzad et al., 2017). These results emphasize the critical importance of having an accurate predictive model for the gasifiers, as it will significantly impact the consequential results and the analysis of the flowsheet. The subsequent calculations of the hybrid model rely on industrial data for validation, which is typically inaccessible. This is a common limitation found in most of Aspen models in the literature.

4. Conclusion

This study aimed to enhance the prediction accuracy of thermochemical conversion models developed using process flowsheet simulations. To achieve this, an integrated Aspen Plus-ML model for lignocellulosic gasification has been developed. A base case process flowsheet simulation is established using Gibbs Free Energy minimisation and the limitations in accuracy are highlighted. A comparative analysis was conducted between various ML methods, including linear regression, ridge regression, regression trees, random forest, XGBoost, and ANN, to select the most accurate approach with the highest R² values. The XGBoost model surpassed other methods, achieving R² values ranging from 0.84 – 0.94 when predicting CO, CO₂, H₂, and CH₄. The accuracy of XGBoost has been also demonstrated when predicting gas and tar yields with R² values of 0.83 and 0.89, respectively. Moreover, a comparison between Aspen Plus and XGBoost prediction accuracy results indicates that the XGBoost algorithm outperformed Aspen Plus calculations, especially when predicting H₂ and CH₄ at high temperatures. Changes in the gasifier output impact the rest of the mass, energy, and environmental aspects of the flowsheet which consequently influence any economic, energy and environmental analyses implemented in the flowsheet. This underscores the importance of developing

Table 9

Comparison of consequential results of different predictive approaches in terms of mass, and energy balance. Gasification was carried out under temperature = 750 °C, S/B ratio = 1.25 and ER = 0 (Loha et al., 2011).

Variable	XGBoost	RGIBBS	(Loha et al., 2011)
CO composition (vol%)*	19.91	15.66	19.74
CO ₂ composition (vol%)	22.25	24.35	21.52
H ₂ composition (vol%)	53.17	59.14	54.17
CH ₄ composition (vol%)	4.68	0.85	4.57
Gas yield (kg/h)	6647.31	9727.01	–
Tar yield (kg/h)	28.87	~0	–
Char (solid carbon) (vol%)	0.09	0.01	–
Gasifier duty (MJ/h)	–8298	–22,699	–
Tar cracker duty (MJ/h)	1913.87	776.21	–
Gas cleaning heat duty (MJ/h)	544,128	551,840	–
CO ₂ emissions (tCO ₂ /y)	16,759.70	9331.59	–

* The computed gas compositions in this table are out of total flow rate of the gaseous products.

highly accurate model for process modellers to facilitate decision making before the plant deployment stage, ensuring the feasibility of the process and avoiding unnecessary resources consumptions.

This study serves as the initial step in developing accurate integrated models for future research. It paves the way for exploring further improvement opportunities in conventional models using ML techniques. This can be achieved by incorporating additional features such as catalyst type, reactor configuration and residence time when developing the model. Another prospect is shifting the focus towards more advanced ML techniques such as active learning and transfer learning.

CRedit authorship contribution statement

Aban Sakheta: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Thomas Raj:** Data curation. **Richi Nayak:** Writing – review & editing, Methodology, Conceptualization. **Ian O'Hara:** Writing – review & editing, Conceptualization. **Jerome Ramirez:** Writing – review & editing, Visualization, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the data in the Supplementary Material file.

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Supplementary materials

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